
Monocular Depth Estimation Report

Computational Intelligence Lab FS2026, ETH Zurich

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Abstract

Monocular depth estimation aims to predict a dense depth map from a single RGB image, a task made challenging by the fact that depth is only indirectly observable from visual cues such as object scale, perspective, occlusion, and scene layout. We develop a sequence of approaches of increasing complexity and evaluate them under the scale-invariant logarithmic (SiLog) loss: a small U-Net baseline, a dilated U-Net variant, a TinyViT encoder paired with a CNN decoder, a custom TinyDPT, the same model trained with knowledge distillation, a DPT-style architecture built on a pretrained ViT backbone with the same distillation process, and finally a ResNet50-TinyViT Hybrid model. Our experiments show that pretrained transformer features and DPT-style multi-scale fusion are especially effective for monocular depth estimation. The ResNet50-TinyViT Hybrid model is more robust and performs better on the Kaggle benchmark. Additionally, we find that the choice of data augmentation must be carefully made to preserve geometric correspondence between the image and the depth target.

1. Introduction

The task of predicting a dense depth map from a single RGB image is fundamentally ill-posed: infinitely many 3D scenes can project to the same 2D image, so depth cannot be recovered from local pixel intensities alone. A successful model must therefore learn to combine a wide range of monocular cues into a coherent prediction, such as object scale, perspective, texture gradients, occlusion boundaries, and global scene layout. The problem has a direct practical importance in robotics, augmented reality, and autonomous navigation, where 3D structure must be recovered without active depth sensors.

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The baseline provided for the project uses a lightweight convolutional model. While such models are efficient, shallow CNNs have limited receptive fields and can struggle to capture long-range dependencies that determine relative depth across a scene. This limitation has motivated a shift toward transformer-based architectures, which can better account for interactions between distant image regions.

We adapt this line of work to the small-data setting of the project. To isolate which architectural choices matter most, we implement and compare a sequence of models of varying complexities. Our best performing model is a ResNet50-TinyViT Hybrid, representing our more novel architecture. We also explored a model that combines a pretrained ViT backbone, a DPT-inspired multi-scale decoder, and knowledge distillation from a larger teacher model. However, that model was not able to outperform the ResNet50-TinyViT Hybrid on the Kaggle benchmark.

2. Models and Methods

2.1. Data Preprocessing

The dataset consists of RGB images paired with depth maps, with metric depth values from approximately 1 millimeters to 80 meters. Inputs were resized to size 384×384 due to GPU memory limitations, and predicted depth maps were upsampled back to 560×560 for submission. Ground-truth depth was clamped and normalized to the range $[0, 1]$, with a binary validity mask excluding invalid pixels from the loss.

2.2. Data Augmentation

We consider four augmentation variants: random horizontal flipping, color jitter, Gaussian blur, and random crop. Random horizontal flipping and color jitter are applied consistently across all approaches, while Gaussian blur and random crop are discarded. Horizontal flipping is safe in this setting because natural scenes are approximately left-right symmetric and the depth map can be flipped jointly with the image without altering depth values. Similarly, color jitter is a sensible augmentation technique as it forces the model to rely on structural geometry rather than color. By contrast, Gaussian blur is discarded as it can interfere with

the model’s ability to utilize fine-grained texture gradients and sharp occlusion boundaries, which are essential statistical depth cues. Random crop followed by resize changes the apparent field of view and object scale without correspondingly updating the camera intrinsics or depth values. This produces physically inconsistent training samples.

2.3. Loss Function

We trained our models using the Scale-Invariant Log (SiLog) loss, which is the standard objective for monocular depth estimation. For each pixel i with valid ground truth, let

$$d_i = \log(\hat{y}_i) - \log(y_i),$$

where $\hat{y}_i$ is the predicted depth and y_i is the ground-truth depth. The SiLog loss is then

$$\mathcal{L}_{\text{SiLog}} = \frac{1}{n} \sum_i d_i^2 - \lambda \left(\frac{1}{n} \sum_i d_i \right)^2$$

, where λ controls the strength of scale invariance. In general, this loss penalizes relative errors and is less sensitive to global scale shifts compared to ordinary pixel-wise regression losses.

2.4. Baseline Tiny U-Net

Our baseline is a lightweight U-Net (Ronneberger et al., 2015) model with an encoder consisting of two downsampling blocks and a bottleneck block at the lowest resolution. The decoder mirrors this structure with two upsampling blocks that use transposed convolutions, each concatenated with the corresponding encoder feature map via a skip connection. A final convolution produces a single-channel output, passed through a sigmoid activation to normalize depth values.

This model served primarily as a sanity check on the data pipeline and training loop. Its receptive field is comparatively limited, so it cannot easily reason about global scene structure, further motivating the transformer-based architectures introduced below.

2.5. Dilated Tiny U-Net

Our second model is a minimal extension of the baseline: we replace the two convolutions in the U-Net bottleneck with dilated convolutions (Yu and Koltun, 2016). These sample the input with gaps between sampled positions, so a 3×3 kernel at dilation 2 has an effective receptive field of 5×5 while keeping the same parameter count. Applied at the bottleneck, where spatial resolution is already low, this enlarges the contextual window that each output pixel sees without adding parameters or compute. The model serves as a controlled test of whether a broader context alone, without

a fundamental change in architecture, is enough to improve depth predictions.

2.6. TinyViT Encoder with CNN Decoder

Next, we built a small vision transformer encoder (Wu et al., 2022) paired with a CNN decoder. The encoder splits the input into patches, projects each patch to a token, adds learned positional embeddings, and processes the tokens through transformer encoder blocks. The token sequence is then reshaped back into a 2D feature map and upsampled to the input resolution by a small CNN decoder.

This was the first model in our progression with a global receptive field at every layer, a property that self-attention provides directly. The trade-off is that ViTs are known to underperform CNNs when trained from scratch on small datasets and can become data-hungry and computationally expensive. This limitation motivates both the distillation experiments and the use of a pretrained backbone described in the following sections.

2.7. TinyDPT

To improve dense prediction quality, we implement a lightweight version of the Dense Prediction Transformer (DPT) (Ranftl et al., 2021). Rather than using only the final encoder output, DPT extracts features from four transformer layers, reshapes them into spatial feature maps through re-assemble blocks and progressively combines them top-down using DPT-style feature fusion blocks. A final depth head with sigmoid activation produces predictions in $[0, 1]$. The architecture consists of four main components:

This design combines the global receptive field of the transformer encoder with the multi-scale spatial recovery characteristic of U-Net-style decoders.

2.8. Knowledge Distillation

We further improve the TinyDPT model by distilling knowledge from a large pretrained depth estimation model (Wang et al., 2021). We use Depth Anything V2 Small as the teacher (Yang et al., 2024), a model trained on tens of millions of images that provides far richer dense supervision than what can be learned from our image training set alone. The student was trained with a combined objective over valid pixels: the standard SiLog loss against the ground-truth and a mean L1 distance against the teacher’s prediction:

$$\mathcal{L} = \lambda_{\text{gt}} \mathcal{L}_{\text{SiLog}}(\hat{y}, y; M) + \lambda_{\text{kd}} \cdot \frac{1}{|M|} \sum_{i \in M} |\hat{y}_i - y_{T,i}|$$

, where $\hat{y}$ is the student prediction, y the ground-truth, y_T

the teacher prediction, and M the set of pixels with valid ground-truth. Because the teacher output is on a different scale than our normalized target, we applied per-image min-max normalization to the teacher’s prediction to $[0, 1]$ before computing the distillation term.

The distillation term gives the student dense supervision over the entire valid region of every image, effectively transferring the visual priors the teacher learned from a much larger training corpus. In our final configuration we use $\lambda_{gt} = 1.0$ and $\lambda_{kd} = 0.5$. Earlier comparison runs used $\lambda_{kd} = 0.3$, but the larger weight produced consistently better results while keeping the ground-truth SiLog loss as the dominant supervision.

2.9. Pretrained ViT DPT

This model replaces the from-scratch encoder of TinyDPT with the pretrained backbone of google/vit-base-patch16-384 (ViT), while keeping the same DPT-style decoder.

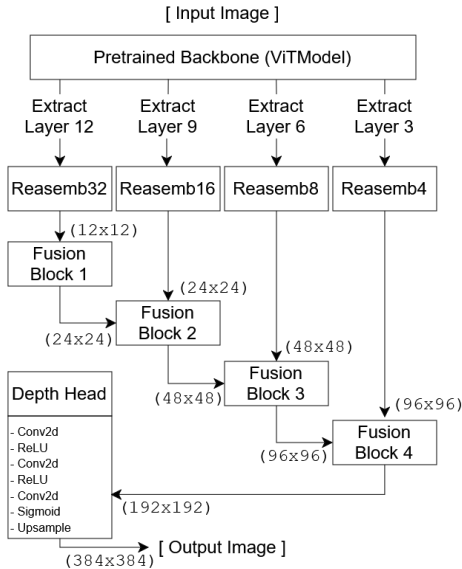


Figure 1. Architecture of Pretrained ViT DPT model.

2.10. ResNet50-TinyDPT Hybrid

As an additional experiment, we implement a hybrid model combining ResNet50 (He et al., 2015) feature extraction with a TinyViT/DPT-style architecture. The motivation is to combine the fact that early convolutional stages efficiently capture local edges and textures, while transformer layers model long-range scene structure. The DPT-style decoder is kept unchanged.

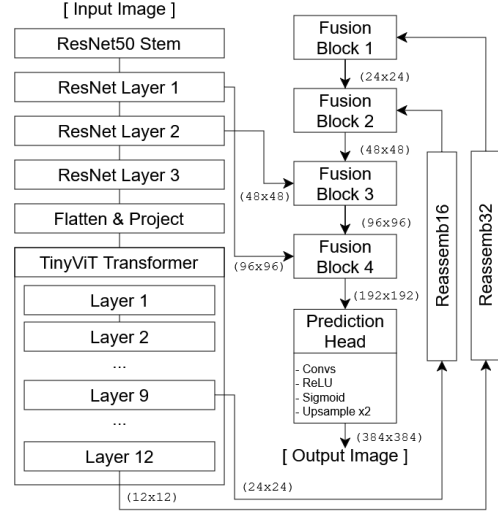


Figure 2. Architecture of ResNet50-TinyDPT Hybrid model.

3. Results

3.1. Experimental Setup

We evaluate all models under a consistent training and evaluation protocol. The training pairs are split into 80% for training (with augmentation), 10% for validation (without augmentation, used for hyperparameter tuning), and 10% for test evaluation. Inputs are resized to 384×384 , and augmentations consist of random horizontal flips (applied jointly to image, depth and mask) together with color jitter on the image only.

All models are trained with the SiLog loss using AdamW and a cosine learning-rate schedule. Mixed-precision training and gradient clipping are applied in all runs for stability and efficiency. The default batch size is 64, except for TinyDPT trained with knowledge distillation, which uses a batch size of 32, and the pretrained ViT-DPT and ResNet50-TinyDPT Hybrid models, which use a batch size of 16. The pretrained ViT-DPT and ResNet50-TinyDPT Hybrid models are trained for 20 epochs using knowledge distillation, while all other models, except for TinyDPT + distillation, are trained for 40 epochs without knowledge distillation. For each run, we retain only the checkpoint with the best validation SiLog, which mitigates overfitting in later epochs. Final results are computed on the test split.

3.2. Main Results

Table 1 reports test SiLog and the corresponding Kaggle scores for all seven models.

While the pretrained ViT DPT achieves the best result with respect to the SiLog loss, the Resnet50-TinyDPT Hybrid performs best on the Kaggle benchmark. The most likely

Table 1. Summary of evaluated approaches.

Model	test_SiLog_loss	Kaggle score
Tiny U-Net	1.288017	0.726
Dilated U-Net	1.183964	0.753
TinyViT + CNN decoder	0.949600	0.621
TinyDPT	0.821655	0.620
TinyDPT + distillation	0.732564	0.622
Pretrained ViT DPT	0.252764	0.544
ResNet50-TinyDPT	0.273424	0.538

explanation is the trade-off between model capacity and overfitting. The pretrained ViT DPT has a huge parameter size and high capacity, which allows it to perfectly minimize the complex SiLog loss. However, this leads to overfitting in the Kaggle benchmark.

On the other hand, the Resnet50-TinyDPT Hybrid acts as a strong regularizer. The convolutional backbone (ResNet50) inherently possesses inductive biases (like translation invariance and locality) that help it generalize better to unseen test data with fewer parameters. This prevents overfitting and results in a higher benchmark score.

Within the four from-scratch models, every architectural step we hypothesized in previous sections reduces the test SiLog loss. Dilation in the U-Net bottleneck improves SiLog from 1.288 to 1.184 (−8%) and replacing the CNN encoder with a transformer takes it to 0.950 (−20%). Adding DPT-style multi-scale fusion reaches 0.822 (−14%) and distillation from Depth Anything V2 brings TinyDPT to 0.733 (−11%). The gains compound monotonically and each step contributes roughly a double-digit percentage. The same ordering of architectural quality is visible between the two pretrained models as well: the pretrained ViT DPT (0.253) outperforms the ResNet50-TinyDPT Hybrid (0.273) by roughly 7.5%, indicating that the choice of architecture continues to matter once a strong pretrained encoder is in place.

However, the largest single transition in the table is from the best from-scratch model to either of the two pretrained models. Incorporating a pretrained encoder reduces test SiLog by a factor of roughly 2.5 to 3, far exceeding any of the individual architectural improvements within each tier. This is consistent with the broader observation in the literature that visual features learned from large-scale image data transfer strongly to dense prediction tasks (Pinheiro et al., 2020).

The Kaggle leaderboard compresses some of these differences, particularly in the middle of the table where TinyViT + CNN decoder, TinyDPT, and TinyDPT + distillation all score around 0.62 despite spanning 0.73 to 0.95 on our test split. The dilated U-Net is a more obvious anomaly: it improves on the test SiLog over the Tiny U-Net but produces

a worse Kaggle score (0.753 to 0.726). These discrepancies suggest that some of the gains observed on our internal test split may not fully generalize to the Kaggle test distribution.

4. Discussion

Our results support three conclusions. First, each architectural step in our progression produces a measurable improvement on the test SiLog, and the same ordering held between the two pretrained models. Second, the largest single jump came from incorporating a pretrained encoder. This is consistent with the well-known data-hungriness of Vision Transformers as our training images are far below the regime in which a from-scratch transformer can learn competitive visual priors (Dosovitskiy et al., 2021). Third, knowledge distillation from Depth Anything V2 provides a smaller but consistent gain, effectively importing supervision derived from a model trained on tens of millions of images.

Our models are trained at 384×384 but evaluated at 560×560 via nearest upsampling, so they are never directly optimized at the submission resolution. The L1 distillation term targets a per-image normalized scale while the SiLog term targets metric-normalized depth, introducing a small inconsistency in the combined objective. The Kaggle scores also compress some of our internal test differences, suggesting a limited transfer of fine architectural gains to the Kaggle environment.

5. Summary

We develop and compare seven models for monocular depth estimation, ranging from a Tiny U-Net baseline through dilated convolutions, transformer encoders, a custom TinyDPT, knowledge distillation from Depth Anything V2, a DPT built on a pretrained ViT-Base backbone, and finally a ResNet50-TinyDPT Hybrid. Our best model, with respect to the Kaggle benchmark, achieves a test SiLog of 0.273 and a Kaggle score of 0.538, comfortably clearing the project 6 baseline. The largest single source of gain is the use of a pretrained encoder, which reduces test SiLog by a factor of 2.5 to 3 relative to the best from-scratch model. However, architectural choices contribute to measurable improvements at every step of the progression, including between the two pretrained models. We also found that augmentations must respect the geometric structure of the depth supervision: horizontal flips and color jitter are safe, while transformations that change the apparent field of view introduce inconsistencies that the scale-invariant loss penalizes sharply.

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